Name: _____ **BME 105 Exam 3**
Student ID:_____

BME 105 Genetics in the Genomics Era
Exam 3, 50 pts

Instructions:

- Read each problem carefully and answer the question in the space provided.
- Write your name on every page of the exam - they will be scanned in and processed by question - we must have your name for you to receive credit.
- You may use a calculator.
- The exam is closed book, closed note. No cell phones, tablets, or laptop computers are allowed to be used. You have the full class time to take the exam.

Chi-square Distribution Table

							Probability							
df	**0.99**	**0.5**	**0.2**	**0.1**	**0.05**	**0.02**	**0.01**	**1.00E-03**	**1.00E-04**	**1.00E-08**	**1.00E-09**	**1.00E-10**	**1.00E-11**	**1.00E-12**
1	0.00	0.45	1.64	2.71	3.84	5.41	6.63	10.83	15.14	32.84	37.32	41.82	46.33	50.84
2	0.02	1.39	3.22	4.61	5.99	7.82	9.21	13.82	18.42	36.84	41.45	46.05	50.66	55.26
3	0.11	2.37	4.64	6.25	7.81	9.84	11.34	16.27	21.11	40.13	44.84	49.54	54.23	58.92
4	0.30	3.36	5.99	7.78	9.49	11.67	13.28	18.47	23.51	43.07	47.88	52.67	57.44	62.20
5	0.55	4.35	7.29	9.24	11.07	13.39	15.09	20.52	25.74	45.79	50.69	55.56	60.41	65.24

Chi-square formula
$X^2 = (E - O)^2 / E$

1) Chromatin constituents and modifications, fill in the blank (6 pts, 1 pt/ea):
Chromatin consists of _____DNA_____, wrapped around ___nucleosomes__, which consist of multiple __histones_____ whose tails can be typically modified by _____acetylation_____ to open chromatin and ____methylation____ to close chromatin. Chromatin accessibility is also modified by DNA _____methylation_____ at CpG sites.

Term bank: DNA, histones, nucleosomes, histone acetylation, histone methylation, cytosine methylation.

2) Epigenetic factors, fill in the blank (4 pts, 1 pt/ea):
The expression of protein-coding genes is regulated by non-coding genes and regions of the genome. Those that are encoded close to the coding gene of interest's location in the genome are termed _____cis___-factors and those that are encoded far away from the coding gene, perhaps on another chromosome, are termed _____trans__-factors. One example of a trans-factor is _____. One example of a cis-factor is __promoter, enhancer, insulator, or CpG island

Term bank: cis, trans, promoter, enhancer, activator, repressor, transcription factor, insulator, CTCF protein, CpG island, DNMT, nucleosome, histone, acetyltransferase

Trans-factors:
activator, repressor,
transcription factor, CTCF protein,
DNMT, nucleosome,
histone, or acetyltransferase

3) Match the sequencing technology with the epigenetic modification it can reveal (4 pts, 1 pt/ea):
   a) Hi-C/Micro-C contact map iii) chromatin loops
   b) ATACseq iv) open chromatin
   c) Bisulfite sequencing ii) methylated cytosines
   d) ChIP-seq i) transcription factor binding sites

   Epigenetic modification options:
   i) transcription factor binding sites
   ii) methylated cytosines
   iii) chromatin loops
   iv) open chromatin

4) Rett syndrome (RTT) is a neurodevelopmental disorder caused by X-linked dominant mutations in the gene encoding methyl-CpG binding protein 2 (MeCP2). MeCP2 is an important reader of DNA methylation. Pathogenic variants in the MeCP2 gene result in the MeCP2 protein being absent or non-functional. Hemizygosity is lethal for males and homozygous females are extremely rare. However, the severity of disease in females heterozygous for the same MeCP2 mutation is highly variable.

   a) What biological process could allow this wide phenotypic heterogeneity to occur (1 pt)?

X-chromosome dosage compensation in females
(Random X-chromosome inactivation (XCI) results in cellular mosaicism in which some cells express wild-type (WT) MECP2 while other cells express mutant MECP2. RTT affects girls due primarily to heterozygous mutations. Only one homozygous female for one allele has been reported.)

   b) What is the probability that a heterozygous female affected with RTT transmits the MeCP2 loss of function allele to her sons (0.5 pt)?

1/2

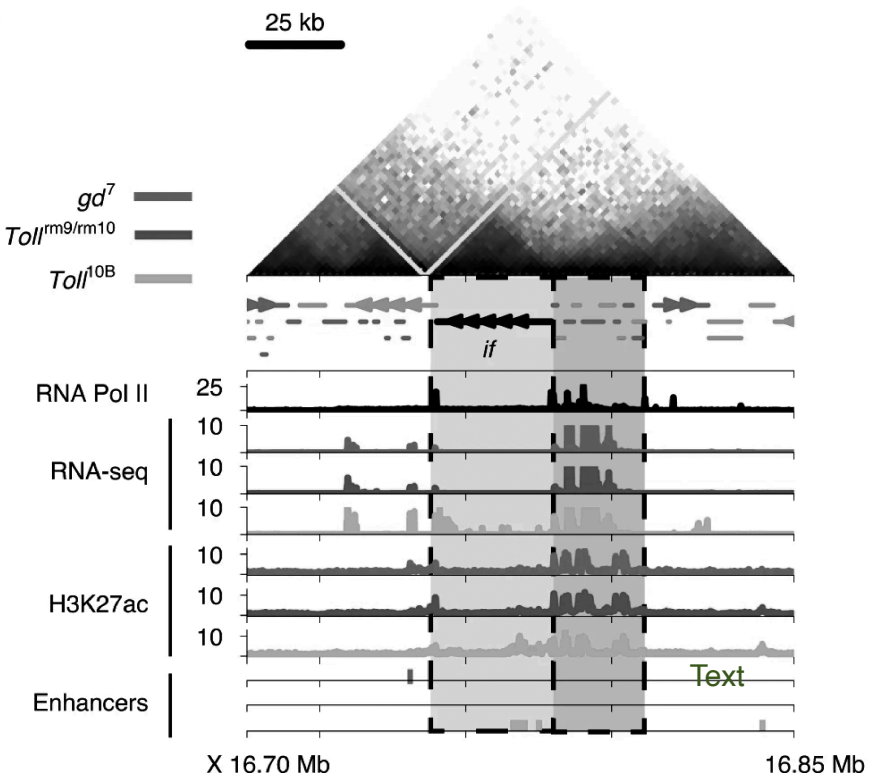
   c) What about her daughters (0.5 pt)?

1/2
If the disease isn't very severe, she's more like a carrier.
But, it is still a dominant trait ... X-inactivation will have to go in her daughter's favor for her to be unaffected

5) What epigenetic inheritance mechanism provides an exception to Mendel's rule that both parents contribute equally to (autosomal) inheritance of traits? Only ~100 genes exhibit this inheritance pattern in humans, and it is exclusive to mammals (2 pt).

Genomic imprinting

6) Use the following example of chromatin organization at the *Drosophila* dorsoventral patterning gene *if* (in black) to answer the questions below. Top, normalized Hi-C contact probability at 2-kb resolution in control embryos at 3–4 hours post fertilization. Positive-strand genes, dark gray; negative-strand genes, light gray. Ser5-phosphorylated RNA Pol II ChIP-seq data (CPM) from control embryos, black. RNA-seq, H3K27ac ChIP-seq data (CPM), and tissue-specific putative enhancers are shown for gd^7 (top), $Toll^{rm9/rm10}$ (middle), and $Toll^{10B}$ (bottom) (5 pts, 1 pt/ea).



a) The "if" region (left light gray-shaded area) is transcribed in all samples:
True or False? _____

b) The region to the right of the "if" region, shaded in dark gray, is transcribed in all samples:
True or False? _____

c) H3K27ac marks are associated with transcription:
True or False? _____

d) What else is associated with H3K27ac besides transcription?
enhancers and RNA Pol II - either works
For the developmental "if" locus, H3K27ac is associated with enhancers.
For the housekeeping locus to the right of "if", H3K27ac is also associated with RNA pol II.

e) What could you conclude about the genes in the light gray shaded region versus the genes in the dark gray region? Which are the housekeeping genes and which are the developmentally regulated genes?

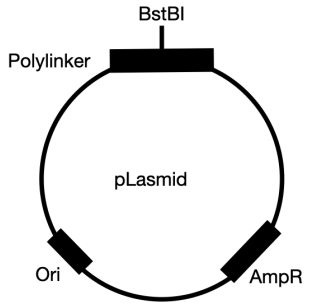
Light gray shaded region:
- ○ Housekeeping genes
- ■ Developmentally regulated genes

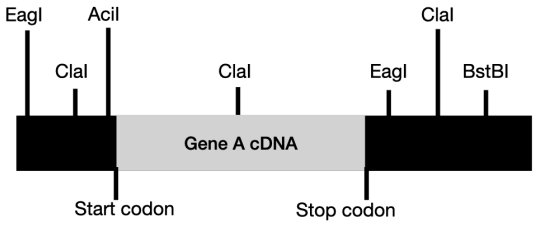
Dark gray shaded region:
- ■ Housekeeping genes
- ○ Developmentally regulated genes

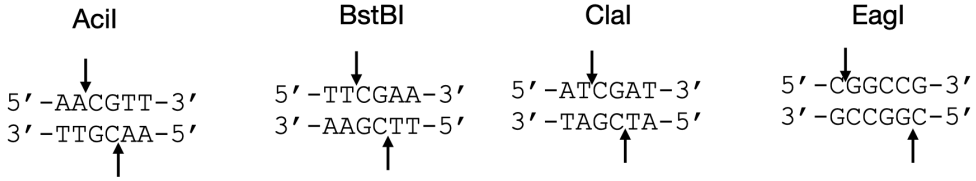
7) You would like to construct a mouse model to study the tumor suppressor gene, *PTEN*, in lung cancer. The cancer mutation is a recessive loss-of-function allele. You first try to make a knockout mouse but homozygous knockout mice die during embryogensis. Describe your follow-up experiment to make a transgenic mouse that would be a good mouse model for the *PTEN* cancer mutation. Describe the linear DNA that will be delivered in the transgene. For example, is there a particular enhancer, promoter, or gene sequence that you would introduce? You may explain in words or pictures (3pts)

lung-specific enhancer is used
the gene sequence is a miRNA targeting PTEN

8) You seek to clone the cDNA of gene A into a plasmid. Which of the following enzyme(s) would you use to digest the linear piece of DNA containing the Gene A cDNA in order to clone the entire gene into the BstBI site in the plasmid? Choose 1 or more of the following enzymes. (4 pts)







- [x] AciI
- [x] BstBI
- [ ] ClaI
- [ ] EagI

9) You are testing the association of diabetes with a SNP located on chromosome 3 in humans. You use a SNP array to genotype individuals with diabetes and a group of controls. The following is the number of individuals of each genotype:

SNP Genotype	Number of Diabetes Cases	Number of Controls
AA	500	1000
AT	500	1000
TT	750	1000

a) To test the null hypothesis that the SNP association is observed by random chance, you perform a X^2 test and fill out the following 2x2 table. The X^2 values should be given up to 2 decimal places. Show your work in order to receive partial credit. (4 pts)

SNP	Cases	Controls	
A	O= 2*500 + 500 =1,500 AA A(T) E= 4500 / 9500 * 3500 = 1657.9 E = Total A / Total genomes * Total cases X^2= (1500-1657.9)^2 / 1657.9 =15.04	O= 2*1000 + 1000 =3,000 AA A(T) E= 4500 / 9500 * 6000 = 2842.1 E = Total A / Total genomes * Total controls X^2=(3000-2842.1)^2 /2842.1=8.77	Total A= 4500
T	O= 2*750 + 500 =2,000 TT A(T) E= 5000 / 9500 * 3500 = 1842.11 X^2= (2000-1842.11)^2 / 1842.11 = 13.53	O= 2*1000 + 1000 =3,000 TT A(T) E= 5000 / 9500 * 6000 = 3157.90 X^2=(3000-3157.9)^2 /3157.9 = 7.90	Total T= 5000
	Total Cases= 3,500	Total Controls= 6,000	Total genomes= 9500

b) What is the total X^2 value (up to 2 decimal places)? Show your work in order to receive partial credit. (1pt) 45.3

c) What is the p-value of the association? (1pt)

between 1E-10 and 1E-11

d) What is the p-value threshold to consider this SNP significant if you tested a total of 1 million SNPs? You can provide your answer as a fraction. (1pt)

threshold of significance is 0.05/1000000
—> p-value<=0.05

e) Is this SNP significantly associated with diabetes if you tested a total of 1 million SNPs? **Yes or no** (1pt)

10) To create a gene knockout mouse, instead of using thymidine kinase as a selection marker, you decide to use green fluorescent protein (GFP). After introducing your DNA construct into ES cells, you forgot to add neomycin to the culture plate to grow up the ES cells. Hoping you can salvage your experiment, which of the following methods would allow you to select ES cells with your gene of interest knocked out? Assume that ES cells with identical DNA sequence will grow in a colony on the plate such that you can select some cells for DNA analysis and the other cells in the colony can be used to inject into blastocysts. (3pts)

- ● Select cells that are not green, perform PCR with primers that would amplify the neomycin-resistant gene.
- ○ Select green cells, perform PCR with primers that would amplify the neomycin-resistant gene.
- ○ Select cells that are not green, perform PCR with one primer in the neomycin-resistant gene and one primer in GFP such that it amplifies more of the DNA construct inserted into the genome.
- ○ Select green cells, perform PCR with one primer in the neomycin-resistant gene and one primer in GFP such that it amplifies more of the DNA construct inserted into the genome.

11) In *Drosophila*, dosage compensation is achieved by expressing a non-coding RNA called *RNA on the X* (*roX*), although it functions differently than *Xist*. *roX* is only expressed in: (3pts)

- ○ female flies.
- ● male flies

12) In a typical knockout mouse experiment, engineered ES cells from an agouti mouse (gray coat color) are injected into a blastocyst derived from a mouse with blackcoat color. These develop into a chimera mouse. You breed this chimera mouse with a purebred black mouse. What phenotypes would you expect in the progeny? As a reminder, the agouti allele is dominant to black (3pt).

- ○ All chimera mice
- ○ All agouti mice
- ○ All black mice
- ○ Some chimera mice, agouti mice, and black mice
- ○ Some chimera mice, agouti mice
- ○ Some chimera mice, black mice
- ● Some agouti mice, black mice

13) To identify enhancer sequences, an enhancerless GFP reporter construct can be randomly inserted into the genome. If inserted near an active enhancer, the construct's GFP reporter gene will be expressed. If it does not insert near an enhancer, no GFP will be expressed. What type of method is used to deliver this type of construct? (3 pts)

- ● Stable transfection
- ○ Transient transfection
- ○ None of the above